

From Pushdown Cancellation to Defect Geometry on Fuss–Catalan Paths

Correction Signatures, a Catalan Boundary, and an Algebraic Defect Hierarchy

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Abstract

We study a combinatorial structure arising from a local cancellation transition in a deterministic pushdown automaton for the binary counting language

$$L_{2,1} = \{w \in \{0,1\}^* : N_1(w) = 2N_0(w)\}.$$

Within the nonnegative 2-Dyck sector, the transition $0, 0/\varepsilon$ corresponds to down-steps landing at heights $3, 5, 7, \dots$. Motivated by this observation, for $d \geq 2$ we distinguish down-steps of a d -Dyck excursion that land at heights $d+1, 2d+1, \dots$. Rather than recording only their number, we retain their ordered landing heights and normalize them by

$$r_i = \frac{c_i - (d+1)}{d}.$$

For a normalized signature $r = (r_1, \dots, r_m)$ we define

$$\Delta(r) = r_m + \sum_{i=1}^{m-1} \psi(r_i - r_{i+1}), \quad \psi(a) = \begin{cases} 0, & a \leq 1, \\ a, & a \geq 2. \end{cases}$$

Our main theorem shows that $\Delta(r)$ is exactly the minimum defect required to realize r : if a path has n down-steps and $m \geq 1$ corrections, then r is realizable at defect $\delta = n - m - 2$ if and only if $\Delta(r) \leq \delta$. Hence

$$n_{\min}(r) = m + 2 + \Delta(r),$$

and an explicit canonical minimum realization is given. The zero-defect signatures are reversed Dyck area sequences and the maximal-correction paths possess a binary cancellation skeleton. The scalar path enumeration contains an auxiliary kernel whose $d = 2$ specialization is OEIS A371395, while the actual correction distribution is a different rational transform of the same kernel. Finally, if $N_{m,\ell}$ counts normalized signatures of length m and exact intrinsic defect ℓ , then

$$(1-q)S(1+qS) = x(1+S)^2(1+q^2S)$$

for $S(x, q) = \sum_{m \geq 1, \ell \geq 0} N_{m,\ell} x^m q^\ell$, and for fixed $j \geq 0$,

$$N_{m,2j} \sim \frac{m^j}{j!} C_m, \quad N_{m,2j+1} \sim \frac{2j+3}{j!} m^j C_m.$$

Thus a local pushdown cancellation rule motivates a d -independent defect geometry with an exact realization theory, a Catalan boundary, algebraic enumeration, and a pairwise asymptotic hierarchy.

Keywords. deterministic pushdown automata; generalized Dyck paths; Fuss–Catalan paths; Catalan words; correction signatures; algebraic generating functions; singularity analysis.

1 Introduction

Deterministic pushdown automata are usually studied through the languages they recognize, their determinism properties, and the resources required for recognition. Individual stack transitions are often treated as implementation details once correctness of the automaton has been established. The present work arose from asking the opposite question: can the locations of a particular stack cancellation carry an intrinsic combinatorial structure?

Consider

$$L_{2,1} = \{w \in \{0,1\}^* : N_1(w) = 2N_0(w)\}. \quad (1)$$

A deterministic pushdown automaton considered by the author for this language uses a center/left/right organization and a stack encoding the imbalance still to be compensated. Formal redrawings of that construction were subsequently used as the motivating automaton in companion work on completion and residual geometry [4, 5]. The present paper follows a different feature of the same automaton: the left-state cancellation transition

$$0, 0/\varepsilon. \quad (2)$$

Under the path encoding

$$1 \mapsto +1, \quad 0 \mapsto -2, \quad (3)$$

accepted words whose running height is nonnegative become 2-Dyck excursions. Within this nonnegative sector, transition (2) corresponds precisely to down-steps whose landing heights are

$$3, 5, 7, \dots$$

The scope qualification is important: the original DPDA recognizes the full language (1), while the Fuss–Catalan model studied here concerns its nonnegative path sector.

Motivated by the 2:1 case, for $d \geq 2$ we consider d -Dyck excursions with steps $+1$ and $-d$, and call a down-step a correction when its landing height satisfies

$$h_{\text{end}} \equiv 1 \pmod{d}, \quad h_{\text{end}} \geq d + 1. \quad (4)$$

For $d = 2$, this is induced by the motivating pushdown cancellation. For $d > 2$, it is used as a combinatorial generalization; no claim is made that the motivating machine itself implements the generalized rule.

The scalar number of corrections is only the first layer of information. We retain their ordered landing heights

$$(c_1, \dots, c_m)$$

and normalize them by

$$r_i = \frac{c_i - (d + 1)}{d}. \quad (5)$$

The resulting sequence $r = (r_1, \dots, r_m)$ is the central object of the paper. We associate with it the functional

$$\Delta(r) = r_m + \sum_{i=1}^{m-1} \psi(r_i - r_{i+1}), \quad \psi(a) = \begin{cases} 0, & a \leq 1, \\ a, & a \geq 2. \end{cases} \quad (6)$$

The main theorem proves that (6) is not an ad hoc weight: it is exactly the minimum realization defect of the signature.

The zero-defect boundary is Catalan. After reversal, its signatures are ordinary Dyck area sequences; at the path level, the correction nodes form a plane full binary tree. Beyond the boundary, all exact defect layers are encoded by a d -independent cubic generating equation and exhibit a pairwise asymptotic hierarchy.

An intermediate scalar enumeration also produces an auxiliary algebraic kernel. In the motivating case $d = 2$, its coefficient triangle is OEIS A371395 [10]. The A371395 array is not the actual cancellation distribution; it is a kernel distribution from which the latter is obtained by a different rational transform. This distinction was one of the steps that led from scalar cancellation counts to ordered correction signatures.

Main contributions. The paper has three principal results.

- (i) *Exact realization geometry.* In the nonnegative 2-Dyck sector, the motivating DPDA cancellation is identified with a precise landing-height event. Its ordered landing heights lead to normalized correction signatures, and the defect functional Δ is proved to characterize realizability exactly:

$$r \text{ is realizable at defect } \delta \iff \Delta(r) \leq \delta, \quad n_{\min}(r) = m + 2 + \Delta(r).$$

The proof gives an explicit minimum witness with no unintended corrections.

- (ii) *Catalan boundary and lossless coordinates.* The zero-defect signatures are exactly reversed Dyck area sequences. At path level the maximal-correction regime has a plane binary cancellation skeleton, and the ordered normalized signature determines that skeleton uniquely; in particular $Q_d(n, n-2) = dC_{n-2}$ for $n \geq 3$.
- (iii) *Autonomous defect hierarchy.* Exact intrinsic-defect signatures satisfy the d -independent cubic equation

$$(1-q)S(1+qS) = x(1+S)^2(1+q^2S),$$

and, for fixed j ,

$$N_{m,2j} \sim \frac{m^j}{j!} C_m, \quad N_{m,2j+1} \sim \frac{2j+3}{j!} m^j C_m.$$

A supporting scalar enumeration gives an exact correction triangle and a universal kernel transform; for $d = 2$ the auxiliary kernel is OEIS A371395, but it is not the actual correction distribution.

2 From the motivating DPDA to correction geometry

2.1 The motivating automaton

The starting point is the author's motivating deterministic pushdown construction for (1). Figure 1 reproduces the formal redrawing used in the companion manuscripts [4, 5]. The full machine is shown to preserve the lineage of the construction, although only one ordinary recognition transition is used to motivate the present combinatorics.

Operationally, the transition

$$0, 0/\varepsilon$$

consumes an input 0 while removing an exposed stack symbol 0. Earlier work used the same automaton to investigate completion information encoded jointly by control state and stack [4]. Here we ask a different question: rather than studying what completion remains after a prefix, we record the moments at which the cancellation transition occurs.

The distinction is fundamental. Membership in (1) depends only on the final balance, while the ordered cancellation history contains information about the internal route by which the computation processes that balance.

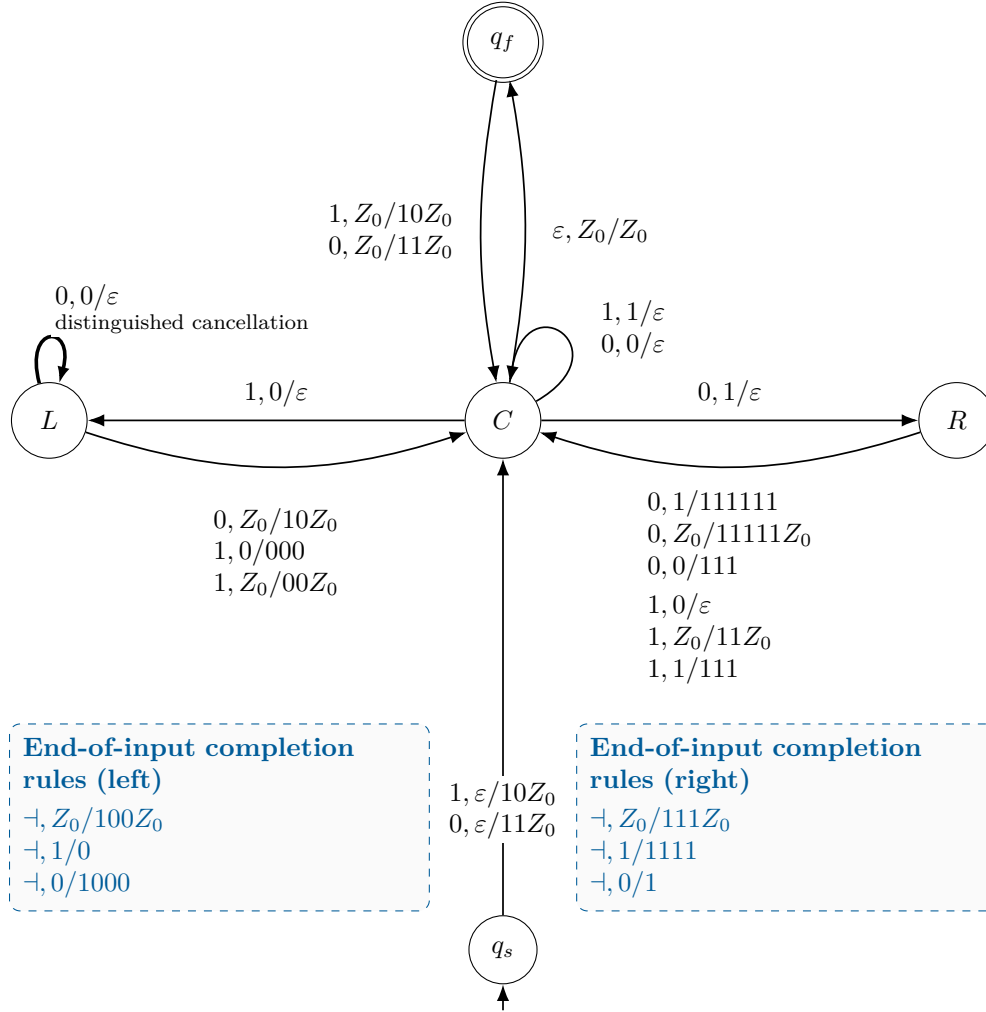


Figure 1: Formal redrawing of the motivating 2:1 DPDA. The heavier left-state loop is the distinguished cancellation $0, 0/\varepsilon$ studied in this paper. The other solid arcs are ordinary recognition transitions; dashed callouts are right-endmarker completion operations and play no role in the correction statistic.

2.2 The nonnegative path sector

For a word $w = w_1 \cdots w_N$, define its running height

$$h(w) = N_1(w) - 2N_0(w),$$

which is exactly the height obtained from (3). If all prefix heights are nonnegative and the final height is 0, then w corresponds to a 2-Dyck excursion.

For completeness, we record the two core invariants needed below directly from the displayed machine. Suppress the permanent bottom marker Z_0 and regard the initial wrapper as entering the semantic core C, L, R with a logical stack in $\{0, 1\}^*$. Write S for this logical stack.

Proposition 2.1 (Reachable-store normal form). *Every reachable core configuration (q, S) satisfies*

$$S \in \mathcal{S}_q, \quad \mathcal{S}_C = 0^* \cup 1^* \cup \{10\}, \quad \mathcal{S}_L = 0^*, \quad \mathcal{S}_R = 1^* \cup \{0\}.$$

In particular, no reachable left-state configuration has logical top symbol 1.

Proof. Induct on the number of consumed input symbols. The two start moves enter the displayed center-store families. The center loops cancel matching exposed symbols and preserve the unary families; the center-to-side mismatch transitions enter only the displayed side families; the left loop $0, 0/\varepsilon$ preserves 0^* ; and every side-to-center transition returns to $0^*, 1^*$, or the exceptional center word 10. Inspection of the solid input-time transitions in Figure 1 shows that no rule enters L with top symbol 1. $\square$

Set $\rho(w) = N_1(w) - 2N_0(w) = h(w)$ and define

$$D(S) = 2N_0(S) - N_1(S), \quad \Phi(C) = 0, \quad \Phi(L) = 3, \quad \Phi(R) = -3.$$

Proposition 2.2 (State-potential invariant). *If a prefix w leaves the core configuration (q, S) , then*

$$\boxed{\rho(w) = D(S) + \Phi(q)}. \tag{7}$$

Proof. For the two wrapper moves, the first consumed symbol 1 enters $(C, 10)$ and gives $D(10) = 1 = \rho(1)$, while the first symbol 0 enters $(C, 11)$ and gives $D(11) = -2 = \rho(0)$. A direct check of each subsequent solid input-time transition in Figure 1 shows that the change in $D(S) + \Phi(q)$ is +1 when the consumed symbol is 1 and -2 when it is 0, exactly the change in ρ . For the distinguished left cancellation $0, 0/\varepsilon$, both sides decrease by 2. $\square$

Corollary 2.3 (Positive odd residuals). *If a reachable core configuration has positive odd residual at least 3, then it is uniquely of the form $(L, 0^k)$ for some $k \geq 0$, and its residual is $2k + 3$.*

Proof. By Proposition 2.1, a center stack is $0^k, 1^k$, or 10, giving residual respectively even nonnegative, nonpositive, or 1. A right-state store is 1^k or 0, giving a negative residual. A left-state store is 0^k , and Proposition 2.2 gives $\rho = 2k + 3$. $\square$

Proposition 2.4 (Motivating $d = 2$ correspondence). *Within the nonnegative 2-Dyck sector of the motivating computation, the transition $0, 0/\varepsilon$ occurs if and only if the corresponding down-step lands at a height in*

$$\{3, 5, 7, \dots\}.$$

Proof. Immediately before a cancellation, the machine is in $(L, 0^k)$ with $k \geq 1$, because a logical top symbol 0 must be present. By (7), the pre-step height is $2k + 3$. Reading the input 0 decreases $h = \rho$ by 2, so the landing height is $2k + 1 \in \{3, 5, 7, \dots\}$.

Conversely, suppose a down-step in the nonnegative sector lands at height $2k + 1 \geq 3$. Its pre-step height is $2k + 3 \geq 5$, a positive odd residual. Corollary 2.3 forces the pre-step core configuration to be $(L, 0^k)$ with $k \geq 1$. Hence the enabled input-0 transition is precisely the left-state cancellation $0, 0/\varepsilon$. $\square$

This proposition is used only to motivate the correction statistic. All combinatorial results from Section 3 onward are stated and proved directly for generalized Dyck paths and do not depend on the remaining transition structure of the automaton.

3 d -Dyck paths, corrections, and signatures

Let $d \geq 2$. A d -Dyck excursion is a word over $\{U, D\}$, where U changes height by $+1$ and D by $-d$, such that every prefix has nonnegative height and the final height is 0. The *size* is the number n of down-steps. Hence a size- n excursion contains dn up-steps.

Definition 3.1 (Correction). A down-step e is a *correction* if

$$h_{\text{end}}(e) \equiv 1 \pmod{d}, \quad h_{\text{end}}(e) \geq d + 1. \quad (8)$$

Equivalently,

$$h_{\text{end}}(e) = d + 1 + dr$$

for a unique $r \in \mathbb{Z}_{\geq 0}$.

Suppose that a path P has $m \geq 1$ corrections, appearing chronologically at landing heights $c_1, \dots, c_m$.

Definition 3.2 (Correction-height signature). The ordered tuple

$$\mathcal{C}(P) = (c_1, \dots, c_m)$$

is the *correction-height signature*. Its normalized form is

$$r(P) = (r_1, \dots, r_m), \quad r_i = \frac{c_i - (d + 1)}{d}. \quad (9)$$

Thus $c_i = d + 1 + dr_i$.

Remark 3.3 (Boundary convention). The signature and defect theory is defined only for paths containing at least one correction, so $m \geq 1$. Correction-free paths remain part of the scalar enumeration $Q_d(n, k)$ in Section 7, but are not assigned a normalized signature or a signature defect.

Definition 3.4 (Path defect). If P has n down-steps and $m \geq 1$ corrections, define

$$\delta(P) = n - m - 2. \quad (10)$$

The nonnegativity of this quantity will follow from the main realization theorem.

Define

$$\psi(a) = \begin{cases} 0, & a \leq 1, \\ a, & a \geq 2. \end{cases} \quad (11)$$

Definition 3.5 (Intrinsic signature defect). For $r = (r_1, \dots, r_m) \in \mathbb{Z}_{\geq 0}^m$, define

$$\Delta(r) = r_m + \sum_{i=1}^{m-1} \psi(r_i - r_{i+1}). \quad (12)$$

The nonlinear threshold in (11) is forced by the local residue geometry, as we now show.

4 Local realization costs

4.1 Gap cost

Suppose two consecutive corrections have normalized heights r and r' , and put

$$a = r - r'.$$

Let b be the number of non-correction down-steps strictly between them and u the number of intervening up-steps. Height balance gives

$$u = d(b + 1 - a). \quad (13)$$

Lemma 4.1 (Local Gap Cost). *For consecutive normalized correction heights r, r' with $a = r - r'$,*

$$b \geq \psi(a).$$

Moreover, equality is attainable.

Proof. If $a \leq 1$, equation (13) permits $b = 0$, with $u = d(1 - a) \geq 0$. Hence

$$U^{d(1-a)}D \quad (14)$$

takes the path directly from correction height $d + 1 + dr$ to $d + 1 + dr'$ and contains no intervening down-step.

Now assume $a \geq 2$. Equation (13) implies $b \geq a - 1$. If $b = a - 1$, then $u = 0$. The first intervening down-step lands at

$$d + 1 + d(r - 1),$$

which is at least $d + 1$ and congruent to 1 (mod d), hence is itself a correction. This contradicts consecutiveness. Therefore $b \geq a$.

Equality is attained by

$$UD^aU^{d-1}D. \quad (15)$$

After the first U , the j th intermediate down-step, $1 \leq j \leq a$, lands at

$$h_j = d + 2 + d(r - j). \quad (16)$$

Thus $h_j \equiv 2 \pmod{d}$; when $d = 2$ this means $h_j \equiv 0 \pmod{2}$. In every case these intermediate down-steps lie outside the correction residue class. After the a th such step the height is $d + 2 + dr'$, the subsequent $d - 1$ up-steps raise it to $2d + 1 + dr'$, and the final D lands at $d + 1 + dr'$. Hence the minimum cost is $a = \psi(a)$. $\square$

4.2 Terminal cost

Lemma 4.2 (Terminal Cost). *If the final correction has normalized height r , then at least $r + 2$ non-correction down-steps are required after it. The bound is sharp.*

Proof. Immediately after the final correction the height is $d + 1 + dr$. If b down-steps and u up-steps remain, then

$$d + 1 + dr + u - db = 0,$$

so

$$u = d(b - r - 1) - 1.$$

Since $u \geq 0$, we have $b \geq r + 2$. Equality is attained by

$$U^{d-1}D^{r+2}. \quad (17)$$

After the up-steps the height is $d(r + 2)$, and the successive down-steps land at $d(r + 1), dr, \dots, d, 0$. All are congruent to 0 (mod d), so none is a correction. $\square$

4.3 Canonical realization blocks

For $r \geq 0$ define the initial block

$$I_r = U^{2d+1+dr} D. \quad (18)$$

Its unique down-step lands at $d + 1 + dr$. For $a \in \mathbb{Z}$, define

$$G_a = \begin{cases} U^{d(1-a)} D, & a \leq 1, \\ U D^a U^{d-1} D, & a \geq 2. \end{cases} \quad (19)$$

For $r \geq 0$ define

$$T_r = U^{d-1} D^{r+2}, \quad P = U^d D. \quad (20)$$

Proposition 4.3 (Block Purity). *Each block has exactly the advertised correction behavior: I_r creates one correction of normalized height r ; G_a creates precisely the next prescribed correction and no unintended correction; T_r creates no correction and returns to height 0; and P is a zero-to-zero padding block with no correction. Whenever a block is started at its prescribed height, all intermediate heights remain nonnegative.*

Proof. The assertions for I_r , G_a , and T_r follow from the landing-height calculations above. In the only potentially dangerous case $a \geq 2$, the intermediate down-steps of G_a have heights (16), outside the correction residue class. The down-steps of T_r and P land at multiples of d . Nonnegativity follows from the explicit minimum heights in the same calculations. $\square$

5 The Defect–Signature Theorem

Let $r = (r_1, \dots, r_m) \in \mathbb{Z}_{\geq 0}^m$, $m \geq 1$, and put $a_i = r_i - r_{i+1}$ for $1 \leq i < m$.

Theorem 5.1 (Defect–Signature Characterization). *For every $d \geq 2$, the following are equivalent:*

- (a) r is the normalized correction signature of a d -Dyck excursion having defect δ ;
- (b) $\Delta(r) \leq \delta$.

Equivalently,

$$\delta_{\min}(r) = \Delta(r), \quad (21)$$

and if r has length m , then

$$n_{\min}(r) = m + 2 + \Delta(r). \quad (22)$$

Proof. Suppose first that a path P realizes r , has n down-steps, and has m corrections. Hence it contains $n - m$ non-correction down-steps. The Local Gap Cost lemma requires at least $\psi(r_i - r_{i+1})$ non-correction down-steps between each consecutive pair of corrections, and the Terminal Cost lemma requires at least $r_m + 2$ after the final correction. Any non-correction down-steps before the first correction only increase the total. Therefore

$$n - m \geq 2 + r_m + \sum_{i=1}^{m-1} \psi(r_i - r_{i+1}) = 2 + \Delta(r),$$

so

$$\delta(P) = n - m - 2 \geq \Delta(r).$$

For sufficiency, let $s \geq 0$ and define

$$W_d(r; s) = I_{r_1} G_{a_1} \cdots G_{a_{m-1}} T_{r_m} P^s. \quad (23)$$

Block Purity shows that this is a nonnegative d -Dyck excursion whose normalized correction signature is exactly r . Its number of down-steps is

$$\begin{aligned} n &= 1 + \sum_{i=1}^{m-1} (1 + \psi(a_i)) + (r_m + 2) + s \\ &= m + 2 + \Delta(r) + s. \end{aligned}$$

Hence

$$\delta(W_d(r; s)) = \Delta(r) + s.$$

Taking $s = 0$ yields a minimum realization. Given any $\delta \geq \Delta(r)$, take $s = \delta - \Delta(r)$. $\square$

Corollary 5.2 (Canonical minimum realization). *Every normalized signature r admits the explicit minimum realization*

$$W_d^{\min}(r) = W_d(r; 0).$$

Here canonical refers to the distinguished block construction (23); no uniqueness claim is made among all minimum realizations of the same signature.

Corollary 5.3 (Signature Persistence). *A signature r is realizable at exactly the defect levels*

$$\Delta(r), \Delta(r) + 1, \Delta(r) + 2, \dots$$

Proof. Append copies of the padding block $P = U^d D$. $\square$

Corollary 5.4 (Maximum number of corrections). *If a size- n d -Dyck excursion has $m \geq 1$ corrections, then*

$$m \leq n - 2.$$

Corollary 5.5 (Zero-defect signatures). *For $r = (r_1, \dots, r_m)$,*

$$\Delta(r) = 0$$

if and only if

$$r_m = 0, \quad r_i - r_{i+1} \leq 1 \quad (1 \leq i < m). \quad (24)$$

After reversal, $a_i = r_{m+1-i}$ satisfies

$$a_1 = 0, \quad a_{i+1} \leq a_i + 1,$$

the usual Dyck area-sequence inequalities. Consequently the number of zero-defect signatures of length m is C_m .

6 What the Defect–Signature Theorem reveals

Theorem 5.1 changes the role of the correction statistic. The original question was forward: given a path, where do corrections occur? The theorem also solves the inverse problem. Given a proposed normalized correction history r , the number $\Delta(r)$ determines exactly the first defect level at which it can occur, and (23) constructs a minimum witness explicitly. Thus

$$\text{path} \longrightarrow \text{signature}$$

is complemented by

$$\text{signature} \longrightarrow \text{minimum realizing path}.$$

In particular, Δ is not merely an enumerative weight. It is the minimum excess over the maximal-correction regime required by the ordered history r . The nonlinear rule in (11) records a

genuine local obstruction: a normalized drop of size at most 1 costs no intervening non-correction down-step, while a drop of size $a \geq 2$ must leave the correction residue class and return to it, incurring exact cost a .

The Catalan family is therefore a boundary rather than the whole theory. The zero set of Δ is precisely the reversed Dyck area-sequence class. Positive defect permits violations of the Catalan nearest-neighbor inequality, but charges every overshoot of size at least 2 by its full magnitude.

After reversal, if $a_i = r_{m+1-i}$, then

$$\Delta^{\leftarrow}(a) = a_1 + \sum_{\substack{1 \leq i < m \\ a_{i+1} - a_i \geq 2}} (a_{i+1} - a_i). \quad (25)$$

Thus the usual 0-based Dyck area sequences are exactly the zero-cost objects for a larger class of nonnegative integer sequences with penalized overshoots; adding 1 to every entry gives the common positive-integer Catalan-word convention.

A second structural feature is universality. The physical correction heights satisfy

$$c_i = d + 1 + dr_i,$$

but r , $\Delta(r)$, the minimum defect, and the realizability criterion contain no d . The parameter d changes the affine embedding of the normalized geometry into path heights, not the intrinsic geometry itself.

For $d = 2$, the automaton supplied the motivating event. After normalization, however, neither the automaton states nor its stack alphabet are needed to define the resulting sequence class. The automaton is therefore the discovery mechanism rather than the final definition of the combinatorial object.

7 Scalar correction enumeration and the A371395 kernel

We now retain only the number of corrections. This scalar enumeration supplies the grammar later used to analyze the maximal-correction skeleton, and it also explains the appearance of OEIS A371395.

7.1 Baseline-shifted excursions and the three-class system

Let $G_h(z, t)$ be the generating function for excursions that start at height h , end at height h , and never go below h . The variable z marks down-steps. A down-step landing at baseline height h carries weight

$$\tau_h = \begin{cases} t, & h \in \{d+1, 2d+1, 3d+1, \dots\}, \\ 1, & \text{otherwise.} \end{cases}$$

The standard first-return decomposition yields

$$G_h = 1 + z\tau_h G_{h+1}G_{h+2} \cdots G_{h+d}G_h. \quad (26)$$

For $h \geq 2$, the marking pattern is periodic with period d , and translation by d preserves the excursion class. Hence

$$G_{h+d} = G_h \quad (h \geq 2). \quad (27)$$

Among the d tail residue classes, let B denote the marked class $h \equiv 1 \pmod{d}$, $h \geq d+1$. Every unmarked tail residue satisfies the same formal equation and hence has the same unique formal-power-series solution; denote it by A . The tail product is $A^{d-1}B$.

For an unmarked tail residue, (26) gives

$$A = 1 + zA^dB. \quad (28)$$

For the marked residue,

$$B = 1 + ztA^{d-1}B^2. \quad (29)$$

Height 1 is exceptional: it lies in residue 1 (mod d) but is not marked. Its equation is

$$G_1 = 1 + zA^{d-1}BG_1,$$

which is the same fixed-point equation as A ; uniqueness therefore gives $G_1 = A$. If $F = G_0$, then $G_1, \dots, G_d$ are all A , and

$$F = 1 + zA^dF. \quad (30)$$

Thus

$$\boxed{F = 1 + zA^dF, \quad A = 1 + zA^dB, \quad B = 1 + ztA^{d-1}B^2.} \quad (31)$$

Define

$$Q_d(n, k) = [z^n t^k]F(z, t). \quad (32)$$

Then $Q_d(n, k)$ counts size- n d -Dyck excursions with exactly k corrections.

7.2 Kernel variable and universal rational transform

Set

$$Y = zA^{d-1}B. \quad (33)$$

From (28) and (29),

$$A = \frac{1}{1-Y}, \quad B = \frac{1}{1-tY}. \quad (34)$$

Hence

$$Y(1-Y)^{d-1}(1-tY) = z. \quad (35)$$

Define the auxiliary kernel series

$$K_d = A^{d-1}B = \frac{Y}{z}. \quad (36)$$

For $n \geq 0$ and $0 \leq k \leq n$, Lagrange inversion gives

$$[z^n t^k]K_d = [z^{n+1} t^k]Y \quad (37)$$

$$= \frac{1}{n+1} [u^n t^k] (1-u)^{-(d-1)(n+1)} (1-tu)^{-(n+1)} \quad (38)$$

$$= \frac{1}{n+1} \binom{n+k}{k} \binom{dn+d-k-2}{n-k}. \quad (39)$$

For $d = 2$, (35) becomes

$$Y(1-Y)(1-tY) = z, \quad (40)$$

and

$$[z^n t^k]K_2 = \frac{1}{n+1} \binom{n+k}{k} \binom{2n-k}{n-k}. \quad (41)$$

This is exactly OEIS A371395 [10], which also records the equivalent series-reversion description associated with $x(1-x)(1-tx)$.

Proposition 7.1 (A371395 kernel identification). *The coefficient triangle of $K_2 = Y/z$ is OEIS A371395.*

The actual correction generating function is different. From (30), (34), and (33),

$$F_d(z, t) = \frac{1 - Y}{1 - 2Y + tY^2}, \quad (42)$$

where Y is determined by (35). Thus

$$\boxed{K_d = \frac{Y}{z}, \quad F_d = \frac{1 - Y}{1 - 2Y + tY^2}.} \quad (43)$$

The rational transform from Y to F_d is independent of d ; all d -dependence lies in the kernel equation.

Proposition 7.2 (Kernel–correction separation). *For $d = 2$, $A371395$ is the coefficient triangle of the auxiliary kernel K_2 , not the actual correction distribution F_2 .*

7.3 Exact correction triangle

Let $U = A - 1$. Eliminating B from (31) gives

$$U(1 + (1 - t)U) = z(1 + U)^{d+1}. \quad (44)$$

Lagrange–Bürmann inversion yields the following exact formula.

Theorem 7.3 (Exact correction triangle). *For $n \geq 1$ and $k \geq 0$,*

$$Q_d(n, k) = \mathbf{1}_{\{k=0\}} + \sum_{j=k+1}^{n-1} \frac{d(n-j)}{j} \binom{j+k-1}{k} \binom{dn-k-1}{j-k-1}. \quad (45)$$

In particular, $Q_d(n, k) = 0$ for $k > n - 2$.

Proof. The coefficient extraction is routine but not central to the signature geometry; a complete Lagrange–Bürmann derivation is given in Appendix A. $\square$

Setting $t = 1$ in (35)–(42) yields the ordinary Fuss–Catalan row sum

$$\sum_{k \geq 0} Q_d(n, k) = \frac{1}{dn+1} \binom{(d+1)n}{n}. \quad (46)$$

Corollary 7.4 (Two boundary columns). *For $n \geq 1$,*

$$Q_d(n, 0) = \frac{1}{n} \sum_{j=0}^{n-1} (n-j) \binom{dn}{j}. \quad (47)$$

For $d = 2$ this simplifies to

$$Q_2(n, 0) = \binom{2n-1}{n-1}. \quad (48)$$

At the opposite boundary, for $n \geq 3$,

$$Q_d(n, n-2) = dC_{n-2}. \quad (49)$$

Proof. These specializations of Theorem 7.3 are recorded in Appendix A; the $d = 2$ zero-correction sum telescopes, while the maximal-correction column contains only the term $j = n - 1$. $\square$

The next section explains the last Catalan factor structurally.
For the motivating $d = 2$ case, the first rows are

$n \backslash k$	0	1	2	3	4	5
1	1					
2	3					
3	10	2				
4	35	16	4			
5	126	94	43	10		
6	462	488	318	132	28	
7	1716	2372	1996	1150	434	84

which differs immediately from A371395. The latter is therefore best viewed as an algebraic waypoint: the scalar cancellation decomposition leads to the A371395 kernel, but the actual correction statistic and the ordered signature geometry are distinct objects.

8 The Catalan boundary and binary cancellation skeletons

The zero-defect regime has $m = n - 2$ corrections. The grammar (31) gives a direct structural explanation of the Catalan factor in (49).

8.1 Residue-refined collapse at maximal correction

Each internal node in the first-return decomposition (26) corresponds to one down-step, and it is marked exactly when its baseline is a correction height. Hence a size- n path with $n - 2$ corrections has exactly two unmarked internal nodes.

The root has baseline 0 and is necessarily unmarked. Before a marked baseline $d + 1$ can be reached, at least one child with baseline in $\{1, \dots, d\}$ must be entered, so at least one further unmarked internal node is necessary. In the maximal-correction regime these are therefore the only two unmarked internal nodes.

Choose the unique nonempty pre-down-step child of the root. If it is slot $j \in [d]$, its baseline is j ; all other root children, including the trailing baseline-0 child, must be empty, since any nonempty one would introduce an additional unmarked internal node. This gives d choices. At the baseline- j node, the unique child that can enter the marked residue without introducing another unmarked node is slot

$$d + 1 - j,$$

whose baseline is $j + (d + 1 - j) = d + 1$. Every other child of this second unmarked node must be empty. Thus all remaining $n - 2$ internal nodes lie in a single marked subtree rooted at correction height $d + 1$.

Now let a marked node have baseline $h \equiv 1 \pmod{d}$, $h \geq d + 1$. Its pre-down-step child baselines are $h + 1, \dots, h + d$, and its trailing child has baseline h . Among these, the only correction baselines are $h + d$ and h , corresponding respectively to slots d and $d + 1$. Since no unmarked internal nodes remain, every other child is empty. Therefore each correction node has exactly two ordered recursive correction slots.

Let $\mathcal{B}_m$ be the set of plane full binary trees with m internal nodes, and let $\mathcal{M}_{d,n}$ be the class of size- n maximally corrected d -Dyck excursions.

Theorem 8.1 (Binary Cancellation Skeleton). *For $n \geq 3$,*

$$\mathcal{M}_{d,n} \cong [d] \times \mathcal{B}_{n-2}. \quad (50)$$

Consequently,

$$Q_d(n, n - 2) = dC_{n-2}.$$

Proof. The forward map records the root entrance slot $j \in [d]$ and then forgets the two forced unmarked entrance nodes, retaining the marked subtree with its two recursive slots d and $d + 1$ as an ordered full binary tree. Conversely, given $j \in [d]$ and a full binary tree, place the first unmarked node in root slot j , place the binary root in its forced slot $d + 1 - j$ at baseline $d + 1$, and recursively realize every binary high/low child in slots $d/d + 1$. All omitted slots are empty. These constructions are inverse. $\square$

The case $n = 2$ is exceptional: there are no corrections and $Q_d(2, 0) = d + 1$.

8.2 Tree coordinates

The standard full $(d + 1)$ -ary tree decomposition of a nonempty d -Dyck excursion is

$$P = UP_1UP_2 \cdots UP_dDP_{d+1}. \quad (51)$$

Assign child weights

$$\omega(j) = j \quad (1 \leq j \leq d), \quad \omega(d + 1) = 0. \quad (52)$$

Lemma 8.2 (Tree-coordinate lemma). *If an internal node v has root-to-node child address $i_1 \cdots i_s$, then the baseline height of its associated down-step is*

$$h(v) = \sum_{\nu=1}^s \omega(i_\nu). \quad (53)$$

In a B -node, the two possible B -children occur in child slots d and $d + 1$, so their baseline increments are respectively d and 0 .

Proof. In (51), the baseline of P_j for $1 \leq j \leq d$ is raised by j , while the suffix P_{d+1} begins again at the parent's baseline after the closing down-step. Iterating along an address gives (53). For a marked baseline $h \equiv 1 \pmod{d}$, among $h + 1, \dots, h + d$ only $h + d$ remains in the marked residue class, while the final child begins at h itself. Hence the two recursive B -children are slots d and $d + 1$. $\square$

Call slot d the *high* binary direction and slot $d + 1$ the *low* direction. If the entrance choice is $j \in [d]$, the binary-root address begins with the two slots $j, d + 1 - j$; by (53) their weights sum to $d + 1$. Thus every entrance choice places the binary root at the same correction height $d + 1$. If $L(v)$ is the number of high edges on the binary-root-to- v path, then

$$c(v) = d + 1 + dL(v), \quad r(v) = L(v). \quad (54)$$

In the traversal order of (51), the slot- d subtree is visited before the node's closing down-step and the slot- $(d + 1)$ subtree after it. Thus correction nodes occur in ordinary binary inorder:

high subtree $\rightarrow$ node $\rightarrow$ low subtree.

If $v_1, \dots, v_m$ are internal binary nodes in inorder, then

$$r_i = L(v_i). \quad (55)$$

8.3 Lossless signatures

Theorem 8.3 (Lossless Signature Theorem). *In the zero-defect regime, the normalized correction signature uniquely determines the binary cancellation skeleton.*

Proof. Let $r = (r_1, \dots, r_m)$ satisfy (24), and let j be the first index with $r_j = 0$. In inorder, the first zero is the binary root. Entries before it lie in the high subtree and each carries one extra high edge, so that subtree has normalized code

$$(r_1 - 1, \dots, r_{j-1} - 1).$$

Entries after the root lie in the low subtree and retain their values,

$$(r_{j+1}, \dots, r_m).$$

Both subcodes satisfy the same zero-defect conditions. Recursive application reconstructs the tree uniquely. $\square$

Hence

binary cancellation skeleton $\longleftrightarrow$ zero-defect signature $\longleftrightarrow$ Dyck area sequence.

The factor d in Theorem 8.1 lies outside this intrinsic Catalan structure; it records only the entrance choice.

If the zero-defect signature has length m , then

$$\sum_{i=1}^m c_i = m(d+1) + d \sum_{i=1}^m r_i. \quad (56)$$

After reversal, $\sum r_i$ is the ordinary area statistic of the corresponding Dyck path. Therefore, with

$$\mathcal{A}_m(v) = \sum_D v^{\text{area}(D)}$$

the Dyck area polynomial,

$$\sum_{P \in \mathcal{M}_{d,m+2}} v^{\sum_i c_i} = d v^{m(d+1)} \mathcal{A}_m(v^d). \quad (57)$$

Thus the Catalan boundary is structural at the level of both counting and correction heights.

9 Algebraic enumeration of defect signatures

We now forget the paths and enumerate the normalized sequences directly. Define

$$N_{m,\ell} = \#\{r \in \mathbb{Z}_{\geq 0}^m : \Delta(r) = \ell\} \quad (58)$$

and

$$S(x, q) = \sum_{m \geq 1} \sum_{\ell \geq 0} N_{m,\ell} x^m q^\ell. \quad (59)$$

Here x marks signature length and q marks exact intrinsic minimum defect.

9.1 Minimum-zero and first-zero decompositions

Let $Z(x, q)$ count nonempty signatures whose minimum entry is 0. Every nonempty signature has a unique decomposition

$$r = \bar{r} + s\mathbf{1}, \quad s = \min_i r_i,$$

with $\min_i \bar{r}_i = 0$. Since translation by $\mathbf{1}$ leaves all adjacent differences unchanged and increases the terminal term by 1,

$$\Delta(r + \mathbf{1}) = \Delta(r) + 1.$$

Therefore

$$Z = (1 - q)S. \quad (60)$$

Let $E(x, q)$ count nonempty signatures ending in 0. For a minimum-zero signature, cut at its first zero. The suffix is arbitrary. For the positive prefix, subtract 1 from every entry. If the reduced prefix is empty there is no contribution; if it ends in 0, the boundary contributes no extra defect; otherwise the boundary contributes one extra unit. Hence the prefix factor is

$$H = 1 + E + q(S - E) = 1 + qS + (1 - q)E. \quad (61)$$

The first-zero decomposition gives

$$Z = xH(1 + S), \quad E = xH(1 + E). \quad (62)$$

Eliminating E and H yields

$$E = \frac{(1 - q)S}{1 + qS}, \quad H = \frac{(1 + S)(1 + q^2S)}{1 + qS}. \quad (63)$$

Theorem 9.1 (Defect–Signature Master Equation). *The formal power series $S(x, q)$ with zero constant term is uniquely determined by*

$$\boxed{(1 - q)S(1 + qS) = x(1 + S)^2(1 + q^2S)}. \quad (64)$$

Equivalently,

$$S = x \frac{(1 + S)^2(1 + q^2S)}{(1 - q)(1 + qS)}. \quad (65)$$

The equation contains no d : the universality of the realization theorem persists at the enumerative level.

9.2 Exact coefficients

Lagrange inversion applied to (65) gives

$$N_{m,\ell} = \frac{1}{m} [u^{m-1} q^\ell] \left(\frac{(1 + u)^2(1 + q^2u)}{(1 - q)(1 + qu)} \right)^m. \quad (66)$$

Expanding the q -dependent factors gives the finite sum

$$\begin{aligned} N_{m,\ell} = \frac{1}{m} \sum_{\substack{a,b \geq 0 \\ a+b \leq m-1 \\ 2a+b \leq \ell}} (-1)^b \binom{m}{a} \binom{m+b-1}{b} \\ \times \binom{2m}{m-1-a-b} \binom{m+\ell-2a-b-1}{\ell-2a-b}. \end{aligned} \quad (67)$$

In particular,

$$N_{m,0} = C_m, \quad (68)$$

$$N_{m,1} = \frac{3m}{m+2} C_m = C_{m+1} - C_m, \quad (69)$$

$$N_{m,2} = \frac{m(m^2 + 10m + 1)}{(m+2)(m+3)} C_m, \quad (70)$$

$$N_{m,3} = \frac{5m(m^3 + 7m^2 + 2m + 2)}{(m+2)(m+3)(m+4)} C_m. \quad (71)$$

The first exact-defect values are

$m \setminus \ell$	0	1	2	3	4	5
1	1	1	1	1	1	1
2	2	3	5	7	9	11
3	5	9	20	35	55	80
4	14	28	76	155	284	476
5	42	90	285	650	1344	2527
6	132	297	1067	2651	6049	12502
7	429	1001	4004	10647	26390	59045

Three arrays should therefore be kept distinct throughout the paper:

K_d (auxiliary kernel counts),

$Q_d(n, k)$ (paths with k corrections),

and

$N_{m,\ell}$ (distinct signatures born at exact defect ℓ).

9.3 Cumulative realizability

Define

$$\hat{N}_{m,\delta} = \#\{r \in \mathbb{Z}_{\geq 0}^m : r \text{ is realizable at defect } \delta\}. \quad (72)$$

By Theorem 5.1 and Corollary 5.3,

$$\hat{N}_{m,\delta} = \sum_{\ell=0}^{\delta} N_{m,\ell}. \quad (73)$$

Thus $N_{m,\ell}$ records the level at which a signature is born, while $\hat{N}_{m,\delta}$ counts all signatures that have appeared by defect δ .

At defect 1, no adjacent drop of size at least 2 is possible. Hence r is realizable at defect 1 if and only if

$$r_i \geq 0, \quad r_i \leq r_{i+1} + 1, \quad r_m \in \{0, 1\}. \quad (74)$$

The map

$$r \mapsto (0, r_m, r_{m-1}, \dots, r_1)$$

is a bijection to Dyck area sequences of length $m + 1$. Therefore

$$\hat{N}_{m,1} = C_{m+1}. \quad (75)$$

Defect 2 is the first layer in which a genuinely non-Catalan local overshoot, such as $r = (2, 0)$, can occur.

10 Asymptotic defect hierarchy

Write

$$S_\ell(x) = [q^\ell]S(x, q) = \sum_{m \geq 1} N_{m,\ell} x^m. \quad (76)$$

Set

$$t = \sqrt{1 - 4x}. \quad (77)$$

At $q = 0$, the master equation gives

$$S_0(x) = C(x) - 1 = \frac{1-t}{1+t}. \quad (78)$$

Proposition 10.1 (Fixed-defect algebraicity and singularity location). *For every fixed $\ell \geq 0$,*

$$S_\ell(x) \in \mathbb{Q}(t), \quad t = \sqrt{1 - 4x}.$$

More precisely,

$$S_\ell(x) \in \mathbb{Q}[t, t^{-1}, (1+t)^{-1}],$$

so on the branch determined by $t(0) = 1$ its only possible dominant singularity is $x = 1/4$.

Proof. Let

$$\mathcal{F}(S, q, x) = (1 - q)S(1 + qS) - x(1 + S)^2(1 + q^2S).$$

The base function $S_0 = (1 - t)/(1 + t)$ belongs to $\mathbb{Q}[t, (1 + t)^{-1}]$. At $q = 0$,

$$\frac{\partial \mathcal{F}}{\partial S} = 1 - 2x(1 + S_0) = t.$$

For $\ell \geq 1$, extraction of q^ℓ makes the unknown S_ℓ occur linearly with coefficient t ; the remaining expression is a polynomial in $x, S_0, \dots, S_{\ell-1}$. Since $x = (1 - t^2)/4$, induction shows that only powers of t and $1 + t$ can occur in the denominator. On the branch $t(0) = 1$, the factor $1 + t$ does not vanish before the square-root branch point $t = 0$, i.e. $x = 1/4$. Moreover, the principal branch $t = \sqrt{1 - 4x}$ is analytic in a slit neighborhood of $x = 1/4$ and $1 + t$ stays nonzero there. Hence every fixed S_ℓ is analytic in a standard dented (Delta-)domain at $1/4$ apart from the explicitly displayed powers of t . This is the local analyticity condition used in the transfer step below. $\square$

10.1 Critical scaling

The scaling expansion below is understood coefficientwise in y . To justify it formally, write $S = 1 + tH$. After substituting $x = (1 - t^2)/4$ and $q = ty$ into the master equation and dividing by t^2 , the resulting equation is analytic near $(t, y, H) = (0, 0, -2)$, and its derivative with respect to H at that point is nonzero. The implicit-function theorem therefore gives a unique expansion $H(t, y) \in \mathbb{Q}[[t, y]]$ on the branch determined by $S(x, 0) = (1 - t)/(1 + t)$.

Introduce

$$q = ty \tag{79}$$

and seek

$$S(x, ty) = 1 + tf(y) + t^2g(y) + O(t^3). \tag{80}$$

Since $S(x, 0) = (1 - t)/(1 + t)$, the correct branch satisfies $f(0) = -2$. Substitution into the master equation gives at order t^2

$$f^2 - 4yf + 8y^2 - 4 = 0, \tag{81}$$

so

$$f(y) = 2y - 2\sqrt{1 - y^2}. \tag{82}$$

At the next order,

$$g(y) = 2 \frac{4y^3 - 3y + \sqrt{1 - y^2}}{\sqrt{1 - y^2}}. \tag{83}$$

For $j \geq 1$,

$$[y^{2j}]f(y) = A_j = \frac{C_{j-1}}{4^{j-1}}, \tag{84}$$

and the odd part of g satisfies

$$g_{\text{odd}}(y) = \frac{8y^3 - 6y}{\sqrt{1 - y^2}} = -6y + \sum_{j \geq 1} B_j y^{2j+1}, \quad B_j = (2j + 3)A_j. \tag{85}$$

Comparing with

$$S(x, ty) = \sum_{\ell \geq 0} t^\ell S_\ell(x) y^\ell$$

yields, for $j \geq 1$,

$$S_{2j}(x) = A_j t^{1-2j} + O(t^{2-2j}), \quad (86)$$

$$S_{2j+1}(x) = B_j t^{1-2j} + O(t^{2-2j}). \quad (87)$$

Using the standard singularity-analysis transfer theorem [7],

$$[x^m](1-4x)^{-\alpha} \sim \frac{4^m m^{\alpha-1}}{\Gamma(\alpha)},$$

and

$$\frac{A_j \sqrt{\pi}}{\Gamma(j - \frac{1}{2})} = \frac{1}{j!},$$

we obtain the main asymptotic result.

Theorem 10.2 (Asymptotic Defect Hierarchy). *For every fixed $j \geq 0$,*

$$N_{m,2j} \sim \frac{m^j}{j!} C_m, \quad (88)$$

and

$$N_{m,2j+1} \sim \frac{2j+3}{j!} m^j C_m. \quad (89)$$

Equivalently,

$$N_{m,2j} \sim \frac{4^m}{j! \sqrt{\pi}} m^{j-3/2}, \quad (90)$$

$$N_{m,2j+1} \sim \frac{2j+3}{j! \sqrt{\pi}} 4^m m^{j-3/2}. \quad (91)$$

Proof. For $j \geq 1$, apply the transfer theorem to (86) and (87). The cases $j = 0$ are boundary cases: $N_{m,0} = C_m$, while

$$N_{m,1} = C_{m+1} - C_m \sim 3C_m.$$

These agree with (88)–(89) at $j = 0$. □

Thus

$$\frac{N_{m,\ell}}{C_m} = \Theta(m^{\lfloor \ell/2 \rfloor}) \quad (\ell \text{ fixed}). \quad (92)$$

The first leading ratios are

$$1, 3, m, 5m, \frac{1}{2}m^2, \frac{7}{2}m^2, \frac{1}{6}m^3, \frac{3}{2}m^3, \dots$$

Hence defect levels occur in asymptotic pairs

$$(0, 1), (2, 3), (4, 5), \dots,$$

and every two defect units introduce one additional power of m relative to the Catalan scale.

This pairing reflects the local geometry: a true violation of the Catalan nearest-neighbor condition requires an adjacent normalized drop of at least 2, and such a drop costs at least two defect units. The factor $m^j/j!$ is consistent with j asymptotically free defect locations among m positions; this is an interpretation of the proven analytic result rather than a separate bijective proof.

For cumulative realizability,

$$\hat{N}_{m,2j} \sim \frac{m^j}{j!} C_m, \quad (93)$$

$$\hat{N}_{m,2j+1} \sim \frac{2j+4}{j!} m^j C_m. \quad (94)$$

At $j = 0$, (94) is consistent with the exact identity $\hat{N}_{m,1} = C_{m+1}$.

11 Relation to existing work

The constructions above intersect several established areas of Catalan and Fuss–Catalan combinatorics. We therefore separate the known ambient frameworks from the correction-signature structure developed here.

11.1 Down-step residues and height-dependent weights

Down-step refinements of generalized Dyck paths by endpoint height modulo the down-step magnitude are already known. Heuberger, Selkirk, and Wagner give exact enumeration formulas for generalized Dyck paths refined by the numbers of down-steps in prescribed height residue classes [8]. Other down-step statistics on generalized Dyck paths have also been studied by Asinowski, Hackl, and Selkirk [1]. Accordingly, the residue condition $h_{\text{end}} \equiv 1 \pmod{d}$ is not itself claimed as new.

The correction statistic also treats the lowest representative $h = 1$ exceptionally. Pétréolle, Sokal, and Zhu develop m -Dyck paths with arbitrary height-dependent weights, including eventually periodic weight sequences [11]. The scalar correction weighting can therefore be viewed as a finite boundary perturbation followed by a periodic tail. We do not claim a new general height-weighted-path framework.

11.2 Catalan and Fuss–Catalan structures

The inequalities

$$a_1 = 0, \quad a_{i+1} \leq a_i + 1$$

are the standard 0-based Dyck area-sequence conditions. Adding 1 entrywise gives the positive-integer Catalan-word convention $w_1 = 1$, $w_{i+1} \leq w_i + 1$ used, for example, in [9]. Catalan words and many of their statistics have an extensive literature; for a recent broad refinement see [3]. The role studied here is not the encoding itself but its occurrence as the exact zero set of the minimum-realization functional Δ . Nor do we claim that the ambient class of nonnegative integer sequences obtained by allowing violations of the Catalan rise condition is new. The contribution is the specific local penalty ψ , its exact interpretation as minimum path-realization defect, and the resulting realization, algebraic, and asymptotic structure.

Likewise, multivariate Fuss–Catalan refinements are classical; Aval’s multivariate Fuss–Catalan numbers provide one representative example [2]. Neither Fuss–Catalan row sums nor multivariate refinement alone is a novelty claim of the present paper.

11.3 A371395 and analytic methods

For $d = 2$, the auxiliary series $K_2 = Y/z$ is precisely OEIS A371395 [10]. This identification is algebraically useful and belongs to the discovery history, but K_2 is not the actual correction distribution and is not the final object generalized by the defect-signature theory.

The uses of Lagrange inversion and singularity analysis are standard. In particular, the coefficient-transfer step in Section 10 is an application of the classical method of Flajolet and Odlyzko [7].

11.4 Position of the present contribution

The paper therefore does not claim as new generalized Dyck paths, height-residue refinement, arbitrary or eventually periodic height weights, Catalan words, Fuss–Catalan refinement, Lagrange inversion, or singularity analysis. The central contribution is instead the package formed by

- ordered correction-height signatures;
- the intrinsic functional Δ and the exact equivalence between $\Delta(r)$ and minimum realization defect;
- the explicit canonical witness construction and signature persistence;
- the zero-defect binary cancellation skeleton and lossless signature code;
- the d -independent cubic master equation for exact intrinsic-defect signatures; and
- the pairwise fixed-defect asymptotic hierarchy.

To the best of our knowledge, this combined correction-signature realization and defect structure has not previously been isolated in this form.

12 Conclusion

We began with a local transition of a deterministic pushdown automaton for the counting language $L_{2,1}$. Within the nonnegative 2-Dyck sector, the transition $0, 0/\varepsilon$ becomes a distinguished landing-height event. Its scalar enumeration leads to a periodic-height grammar, a universal kernel transform, and, for $d = 2$, the auxiliary OEIS triangle A371395.

The main structure emerges only when the cancellation events are retained in chronological order. Their normalized landing heights form correction signatures, and the functional

$$\Delta(r) = r_m + \sum_{i=1}^{m-1} \psi(r_i - r_{i+1})$$

measures exactly the minimum defect needed to realize a prescribed history. The minimum size is $m + 2 + \Delta(r)$, and the proof gives an explicit canonical witness with no unintended corrections.

At zero defect, the correction nodes form a plane binary skeleton, the normalized signature is a lossless coordinate code, and reversal gives an ordinary Dyck area sequence. Beyond this Catalan boundary, the signatures form a universal algebraic hierarchy governed by

$$(1 - q)S(1 + qS) = x(1 + S)^2(1 + q^2S),$$

and fixed defect satisfies

$$N_{m,2j} \sim \frac{m^j}{j!} C_m, \quad N_{m,2j+1} \sim \frac{2j+3}{j!} m^j C_m.$$

Thus the development can be summarized as

pushdown cancellation $\longrightarrow$ distinguished path event $\longrightarrow$ ordered correction signature
 $\longrightarrow$ minimum-defect geometry $\longrightarrow$ Catalan boundary $\longrightarrow$ algebraic and asymptotic hierarchy.

The final combinatorial object no longer requires the automaton for its definition. Nevertheless, the automaton remains essential to the discovery narrative: it supplied the local cancellation rule from which the defect geometry emerged. In this sense, the automaton is not merely an application of the combinatorics; it is the source from which the combinatorial geometry was discovered.

Outlook. The ordered nature of correction signatures suggests finer chronological refinements in which interactions among consecutive correction events are retained rather than collapsed to the single scalar defect Δ . A companion manuscript develops one such third-order arithmetic lift, called a chronological carry refinement, and studies its fixed-column and p -adic geometry [6]. That additional structure is logically separate from the results proved here: the present paper supplies the ordered local correction geometry, while the arithmetic polarization is introduced independently in the companion work.

A Scalar coefficient details

For completeness we give the coefficient calculations omitted from Section 7.

Proof details for Theorem 7.3

From $F = (1 - z(1 + U)^d)^{-1}$,

$$F = \sum_{s \geq 0} z^s (1 + U)^{ds}.$$

Using (44) in Lagrange–Bürmann form and setting $j = n - s$, for $1 \leq j \leq n - 1$ one obtains

$$[z^j](1 + U)^{d(n-j)} = \frac{d(n-j)}{j} [u^{j-1}] \frac{(1+u)^{dn+j-1}}{(1+(1-t)u)^j}.$$

Moreover,

$$[t^k](1 + (1-t)u)^{-j} = \binom{j+k-1}{k} u^k (1+u)^{-j-k},$$

and therefore

$$[u^{j-k-1}](1+u)^{dn-k-1} = \binom{dn-k-1}{j-k-1}.$$

Summing over j gives (45); the constant contribution is $\mathbf{1}_{\{k=0\}}$.

Boundary-column specializations

Putting $k = 0$ in (45) and using

$$\frac{d}{j} \binom{dn-1}{j-1} = \frac{1}{n} \binom{dn}{j}$$

gives (47), including the $j = 0$ term. For $d = 2$,

$$\frac{n-j}{n} \binom{2n}{j} = \binom{2n-1}{j} - \binom{2n-1}{j-1},$$

so the sum telescopes to (48). Finally, setting $k = n - 2$ leaves only $j = n - 1$ and yields

$$Q_d(n, n-2) = \frac{d}{n-1} \binom{2n-4}{n-2} = dC_{n-2}.$$

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